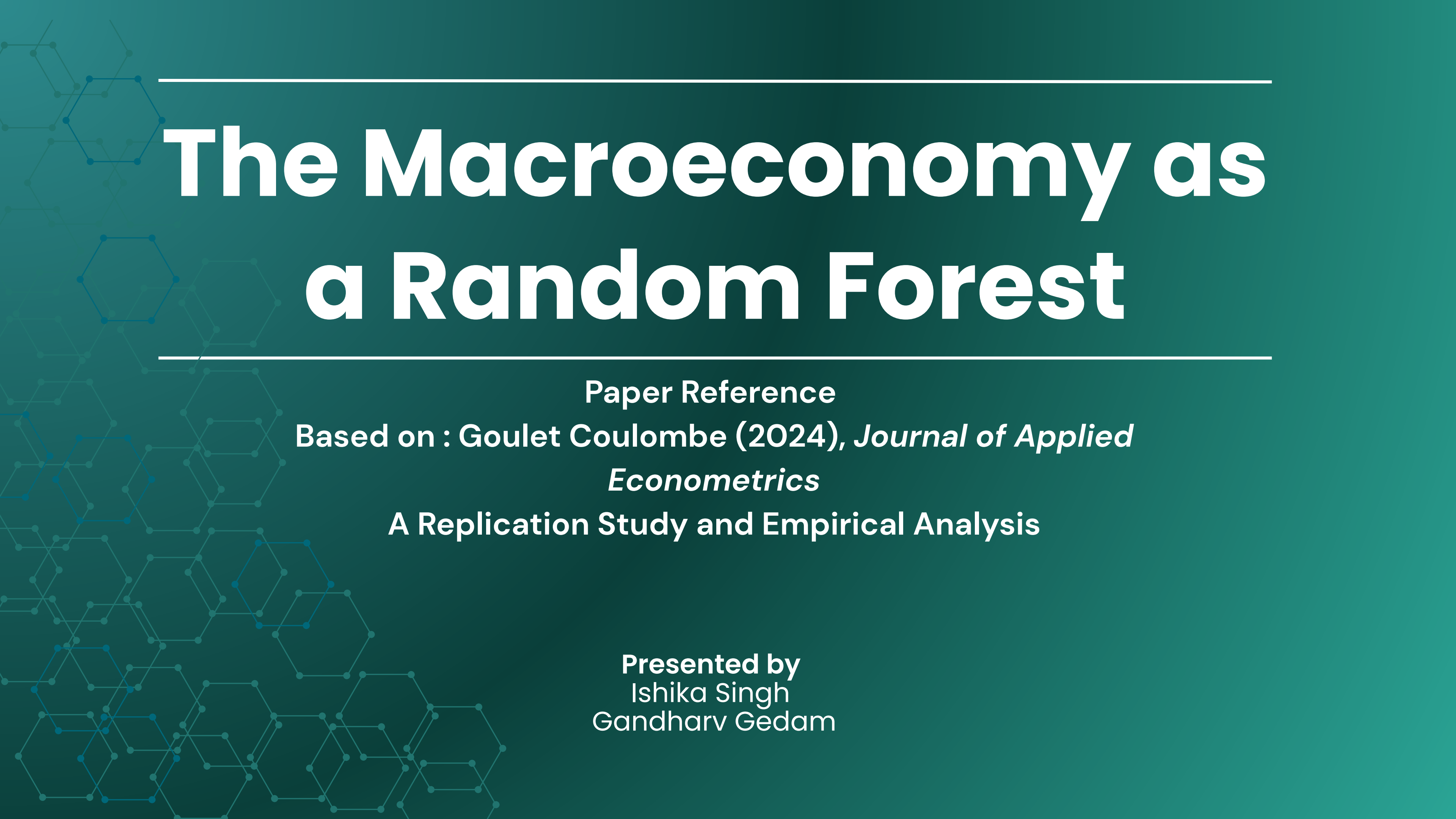




The Macroeconomy as a Random Forest

Paper Reference

Based on : Goulet Coulombe (2024), *Journal of Applied
Econometrics*

A Replication Study and Empirical Analysis

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Motivation : The Forecasting Problem

THE PROBLEM

- Standard linear models (AR, VAR) assume fixed parameters over time
- Economy undergoes structural breaks, regime switches, and crises
- Failed to predict 2008 recession unemployment spike
- Nonlinear models (SETAR, STAR) are rigid and data-hungry
- ML methods lack interpretability for economic theory

THE MRF SOLUTION

- Adapts Random Forest to model time-varying coefficients β_t
- Nests threshold, smooth-transition, and structural break models
- Uses large data (FRED-QD) for state variable selection
- Maintains interpretability via Generalized TVPs (GTVPs)
- Forecasts better while remaining econometrically grounded

MRF MODEL ARCHITECTURE

$$y_t = X_t' \beta_t + \varepsilon_t \quad \text{where} \quad \beta_t = F(s_t)$$

X_t (Linear Part)

Macroeconomic regressors:
AR lags, factor-augmented variables (FA-ARRF)

Holds economic meaning as TVP equation

F (Random Forest)

Ensemble of trees grown on state variables s_t

Captures complex nonlinearities & high-dim data without overfitting

β_t (GTVPs)

Generalized Time-Varying Parameters — the output

Nests: switching, smooth transition, breaks, and random walk TVPs

KEY METHODOLOGICAL INNOVATIONS

Random Walk Regularization

Instead of letting coefficients jump wildly, the model nudges each coefficient toward its neighbours in time ($t-1, t+1$). Economic conditions rarely flip overnight — this keeps estimates realistic and stable

Moving Average Factors (MAFs)

Instead of feeding the forest hundreds of raw variable lags, we first use PCA to compress them into two smooth summary scores. This prevents the forest from wasting splits on repeating lag information and preserves degrees of freedom.

Block Bayesian Bootstrap (BBB)

The coefficient estimate varies slightly among the 500 trees in the forest. To determine whether "this regime shift is statistically meaningful" or "this is just noise," we utilize the distribution of these estimations as error bars.

Surrogate Trees for Interpretability

After getting the complex time-varying coefficients, we fit a tiny 3-leaf decision tree to explain WHAT drives them. For example: ' y_2 doubles when private employment falls below threshold X ' — giving clear economic intuition.

DATA

- FRED-QD: 248 US macro/financial variables, 1960Q1–2014Q4
- Targets: GDP, Unemployment, CPI Inflation
- OOS period: 2003Q1 – 2014Q4 (expanding window)

FORECASTING SETUP

- Horizons: $h = 1, 2, 4$ quarters ahead
- Method: Direct forecasts; RMSE evaluation
- Benchmark: AR(4) model; DM test for significance

MODELS COMPARED IN REPLICATION

Benchmarks

AR(4), FA-AR,
Lasso,Ridge

MRF Variants

ARRF, FA-ARRF,

Classical NL

SETAR

TVP Models

TV-AR, TV-FAAR

DATA PREPRATION

HOW WE USED IT

- Downloaded from FRED St. Louis website
- Applied the TCODE transformation to each series to achieve stationarity
- Series not ending in 'x': taken directly from FRED database
- Series ending in 'x': manually modified (e.g. PCDGx deflated by own price index)
- Monthly series aggregated to quarterly via averaging
- S&P 500 data sourced from public datasets as noted in the paper

TRANSFORMATION CODES (TCODE)

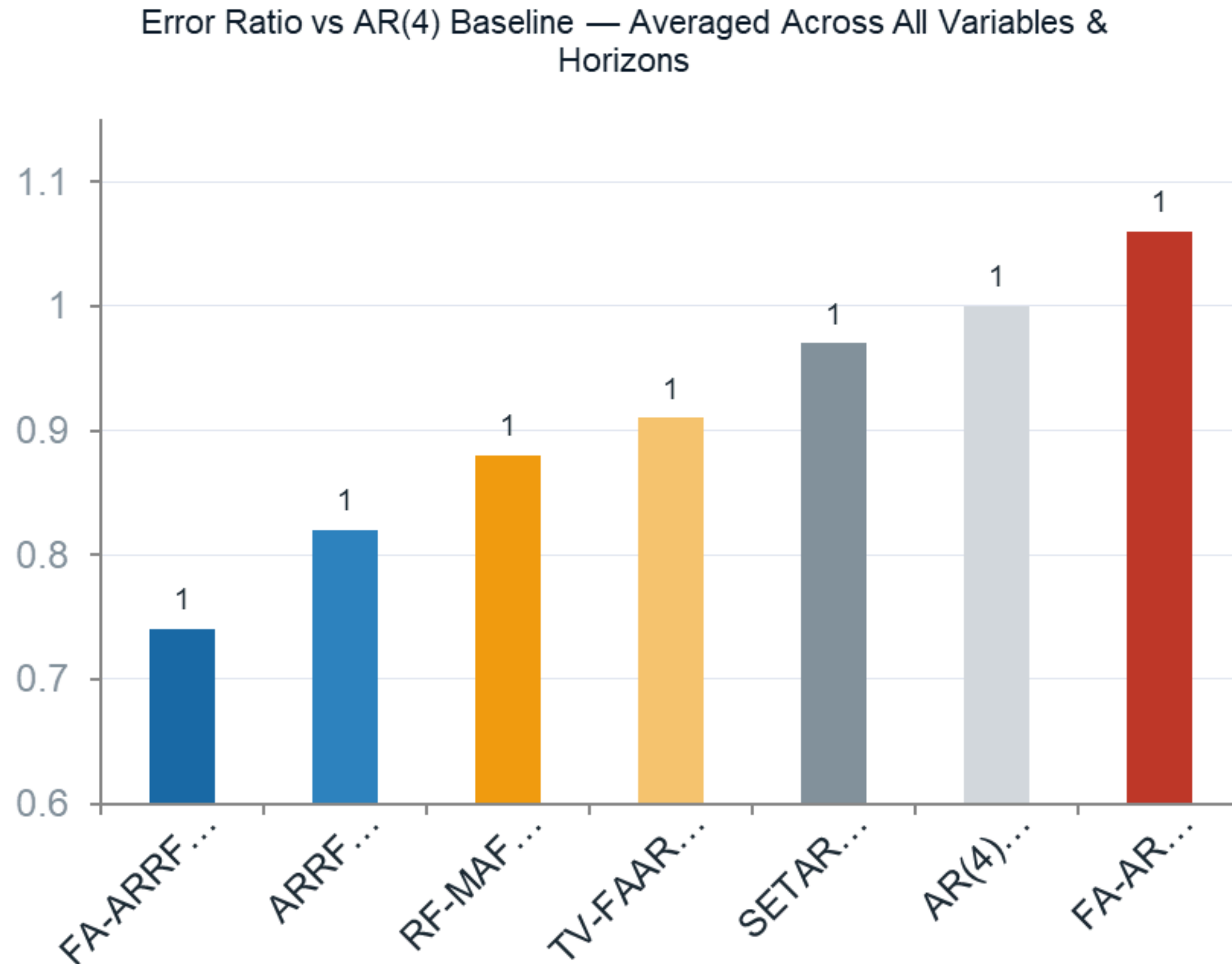
- 1 — No transformation (level)
- 2 — First difference: Δx_t
- 3 — Second difference: $\Delta^2 x_t$
- 4 — Log level: $\log(x_t)$
- 5 — Log first difference: $\Delta \log(x_t)$ ← most common (growth rates)
- 6 — Log second difference: $\Delta^2 \log(x_t)$
- 7 — Percent change: $\Delta(x_t/x_{t-1} - 1)$

Engineering The State Variable S_t

S_t is the information set available to the forest for determining β_t — its design is crucial for performance

Component	Purpose	Construction
8 lags of y_t	Self-exciting AR dynamics — e.g. rising unemployment predicts more rises	<i>Used directly</i>
Time trend t	Captures slow, long-run structural change — like shifting inflation expectations over decades	<i>Counter Variable (1, 2, 3...)</i>
2 lags of all FRED-QD vars	Allows fast regime detection — a sudden credit crunch or yield curve inversion triggers a switch	<i>Raw FRED-QD with TCODE applied</i>
8 lags of 5 PCA factors F	Compresses the whole economy into 5 summary dimensions (growth, financial stress, etc.)	<i>Standard PCA across all FRED-QD series</i>
2 MAFs per variable j	Tells the forest the recent trend of each indicator without using up lots of tree branches	<i>PCA applied to panel of 8 lags of variable j</i>

MAIN FORECASTING RESULTS (RMSE/RMSE_AR)

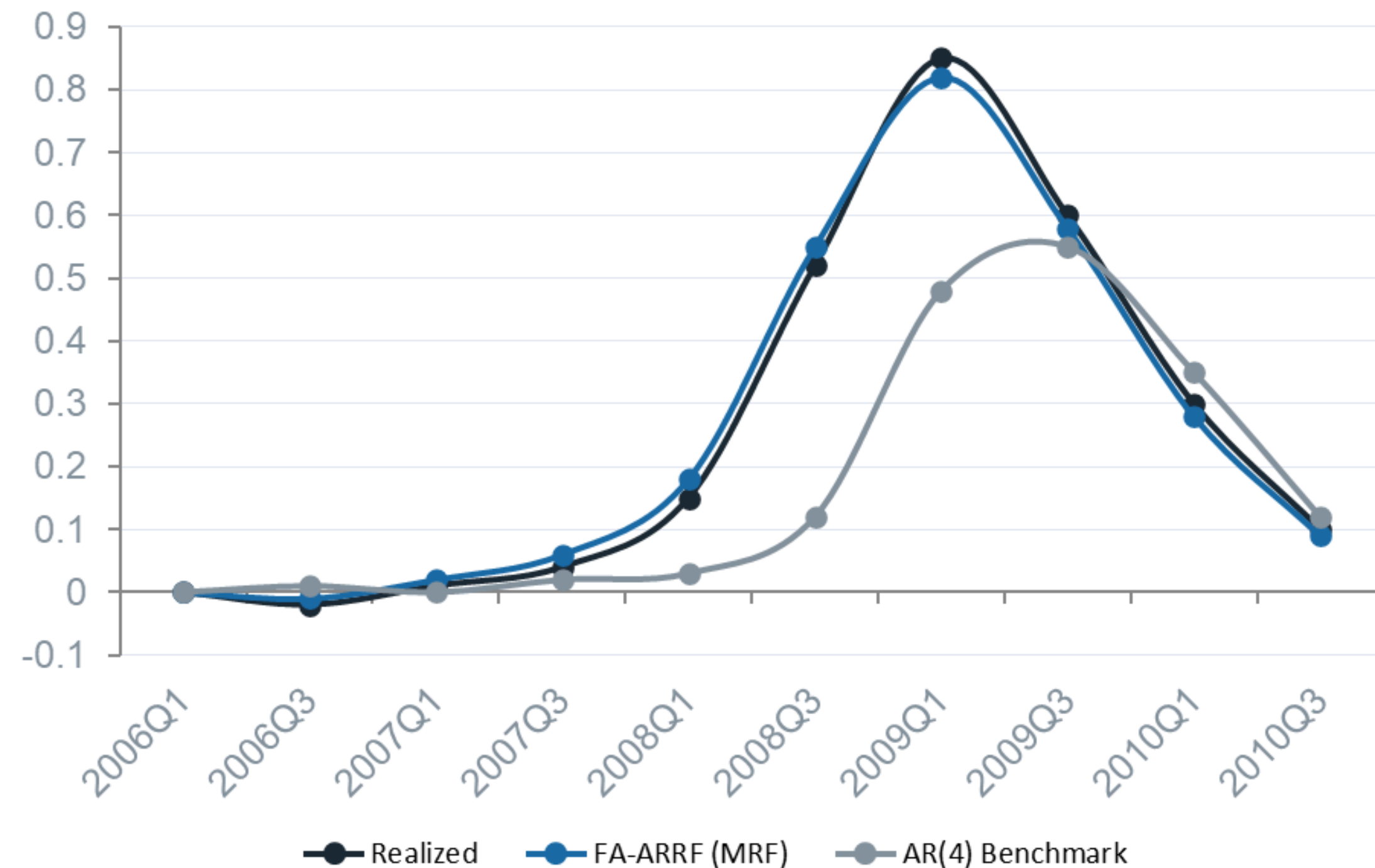


KEY FINDINGS

- **FA-ARRF achieves ~26% RMSE reduction for unemployment over AR(4)**
- **MRF variants outperform all traditional nonlinear models (SETAR, STAR)**
- **MAFs add ~15% gain over raw RF alone**
- **FA-AR fails despite rich data — linear part misspecification**
- **MRF forecasts 2008 recession 1–2 quarters ahead of standard models**

PREDICTING THE 2008 UNEMPLOYMENT SPIKE

Unemployment Change Forecast (h=1)



WHY MRF WINS

- MRF explains 80% of unemployment variation using only data available before the crisis
- 26–30% less error than AR across all horizons 1 through 8 quarters
- Term spreads, building permits, inventories — their coefficient **DOUBLES** as the recession approaches
- Model trained only on data up to 2007Q2 — this is a genuine real-time forecast
- Standard model (AR) only reacted after unemployment had already surged

GTVP ANALYSIS: UNEMPLOYMENT EQUATION

FA-ARRF equation: $y_{t+1} = \mu_t + \varphi_{1,t} y_t + \varphi_{2,t} y_{t-1} + \gamma_{1,t} F1_t + \gamma_{2,t} F2_t + u_{t+1}$

Intercept μ_t

Alternates between ≥ 2 regimes — positive trend entering recessions, slight negative trend during expansions

Persistence $\varphi_{1,t} + \varphi_{2,t}$

Strikingly time-invariant — comparable to OLS. MRF learns persistence is stable; variation is elsewhere

Real Activity Factor $\gamma_{1,t}$

Roughly within OLS bands; doubles around recessions but with high uncertainty — not the main driver

Forward-Looking Factor $\gamma_{2,t}$

KEY DRIVER: Smooth switching — nearly doubles before every recession, including OOS in 2008. Correlated with spreads, permits, capacity utilization

Phillips Curve : Not Dead but Cyclical

$$\pi_t = \mu_t + \beta_{\{1,t\}} \hat{\pi}^{\text{SR}}_t + \beta_{\{2,t\}} u^{\text{GAP}}_t + \beta_{\{3,t\}} \pi^{\text{IMP}}_t + \varepsilon_t$$

$\beta_{\{1,t\}}$: Short-Run Expectations Weight

- **Secular decline:** Steadily decreasing since 1983 as inflation became more anchored
- **Cyclical pattern:** Rises during expansions, collapses in every recession, then gradually recovers
- **Mechanism:** Post-recession, long-run expectations are more reliable; as expansion ages, short-run lags become more informative again

$\beta_{\{2,t\}}$: Unemployment Gap Coefficient

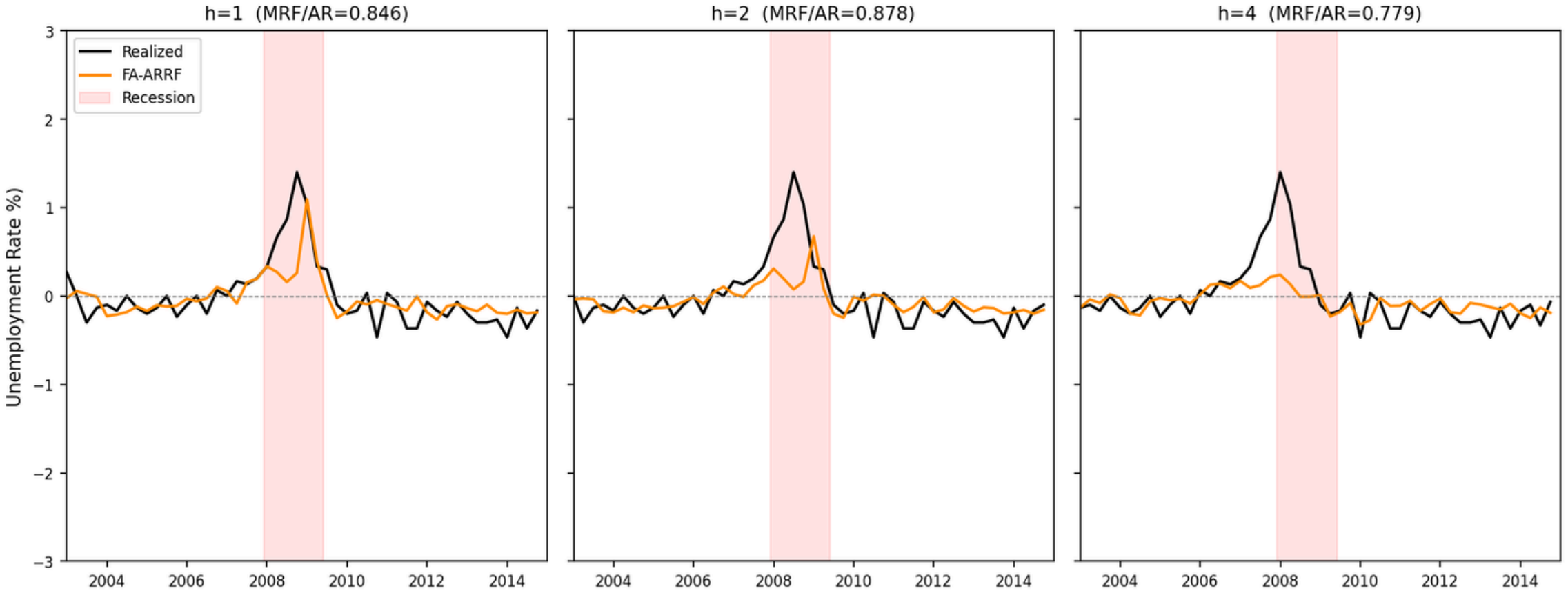
- **Long-run flattening:** Significant decline from 1980s – confirms Blanchard et al. (2015)
- **But also cyclical:** $\beta_{\{2,t\}} \rightarrow 0$ following EVERY recession since 1960, not just 2008
- **Convex PC:** Strong response to positive unemployment gap (overheating) but muted to slack – $\text{corr}(\beta_{\{2,t\}}, \text{TCU}) = 0.81$

WHY GTVPs ARE BETTER THAN TRADITIONAL TVPs

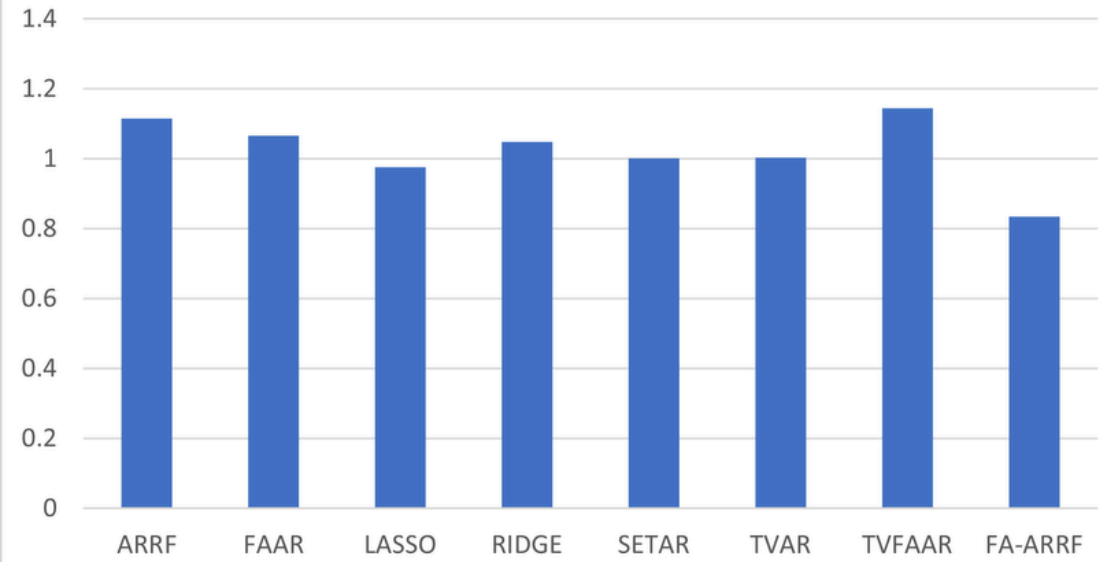
<u>Traditional Random Walk TVPs</u>	<u>Generalized TVPs from MRF</u>
• Only uses time — if two periods are close in time, they get similar coefficients	• Uses 248 economic variables — if two periods look economically similar, they share coefficients
• Designed for slow, continuous drift — terrible at sudden regime switches	• Flexible: can do slow drift AND sudden switching AND cyclical patterns all at once
• Forecast problem: at the edge of the sample, estimates are unstable and change a lot as new data arrives	• No edge problem — the forest reads current conditions from S_t and projects forward cleanly
• Result: coefficients often barely move from OLS estimates (flat and boring)	• Coefficients show sharp, statistically significant shifts around recessions and turning points
• More variation forces higher variance; less variation misses the actual regimes	• Forest diversity keeps variance low while still capturing the true dynamics

UNEMPLOYMENT FORECASTING RESULTS

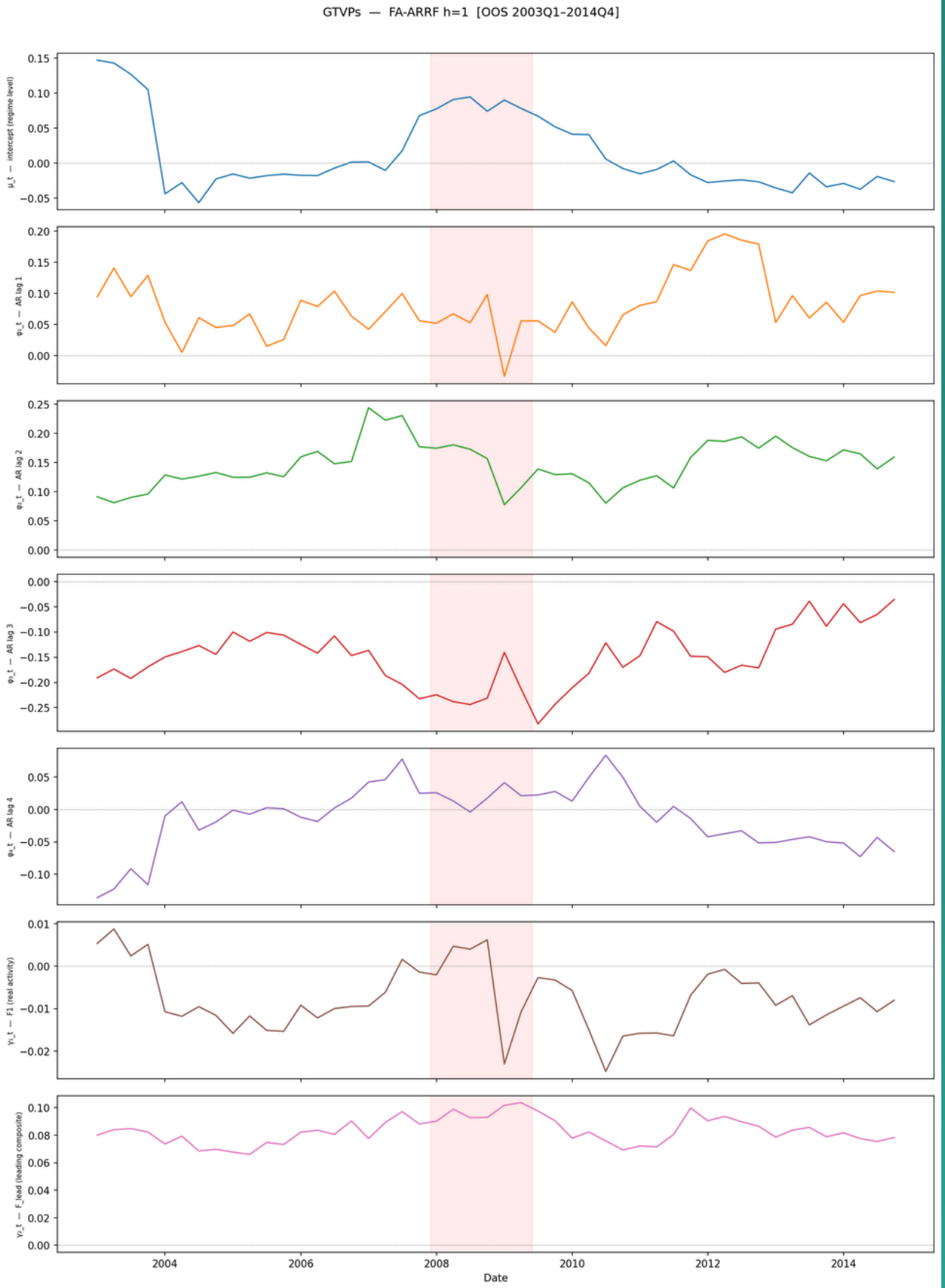
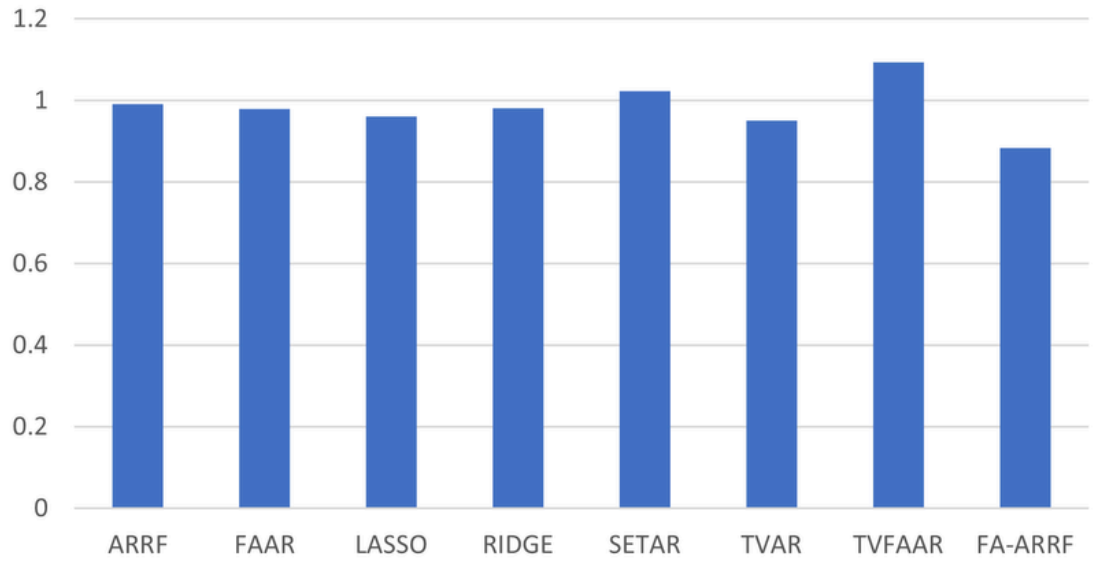
FA-ARRF Unemployment Rate forecasts vs Realized



RELATIVE RMSE(UR,H=1)

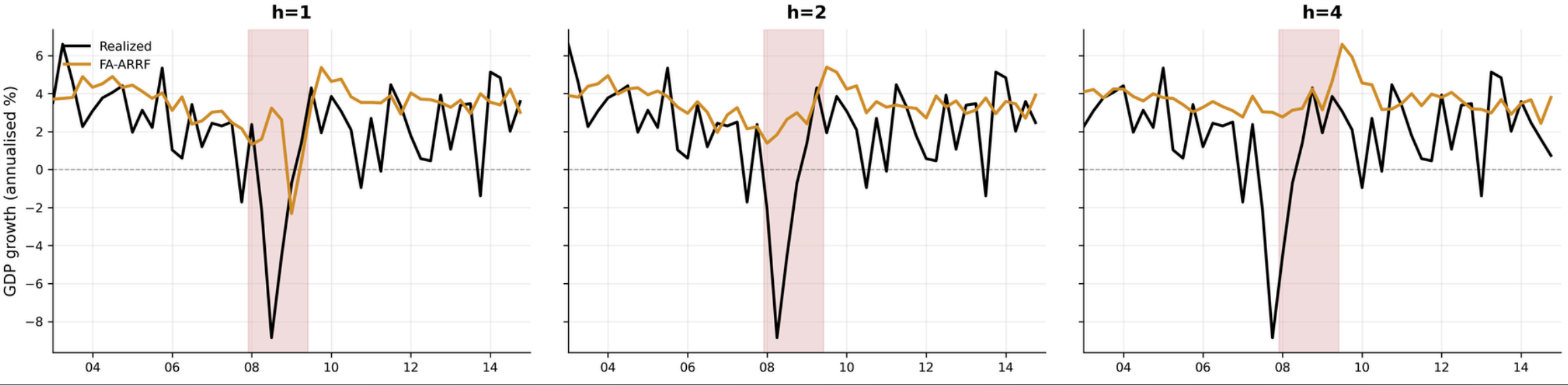


RELATIVE RMSE(UR,H=2)

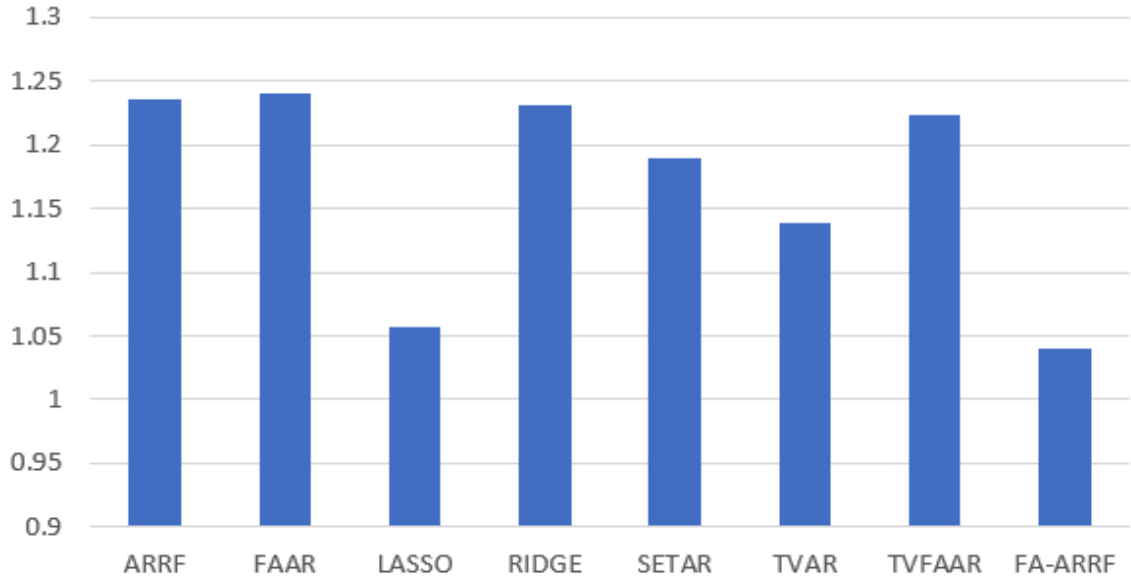


GDP FORECASTING RESULTS

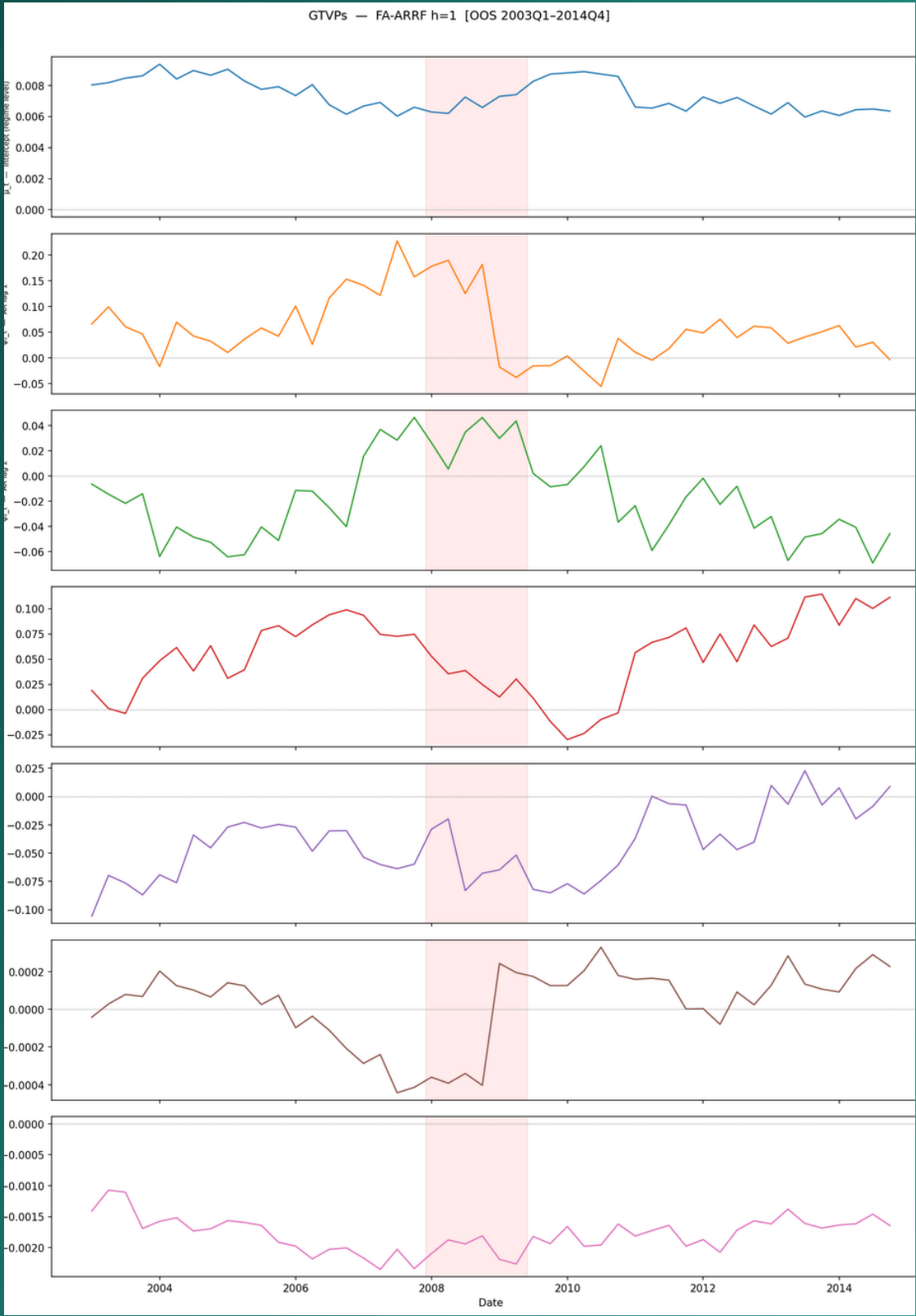
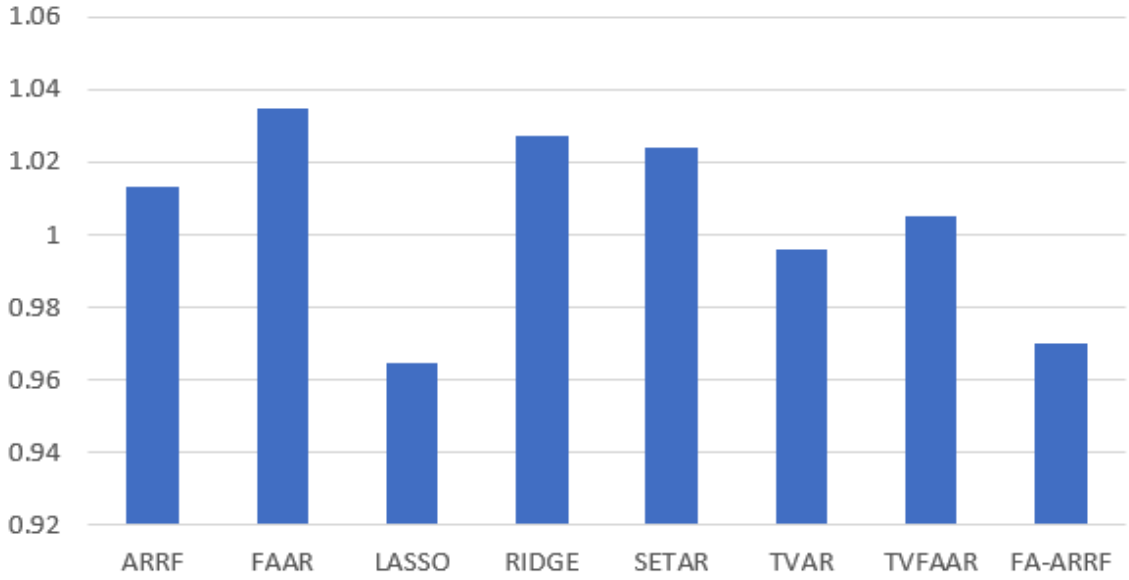
FA-ARRF GDP forecasts vs Realized [2003Q1-2014Q4]



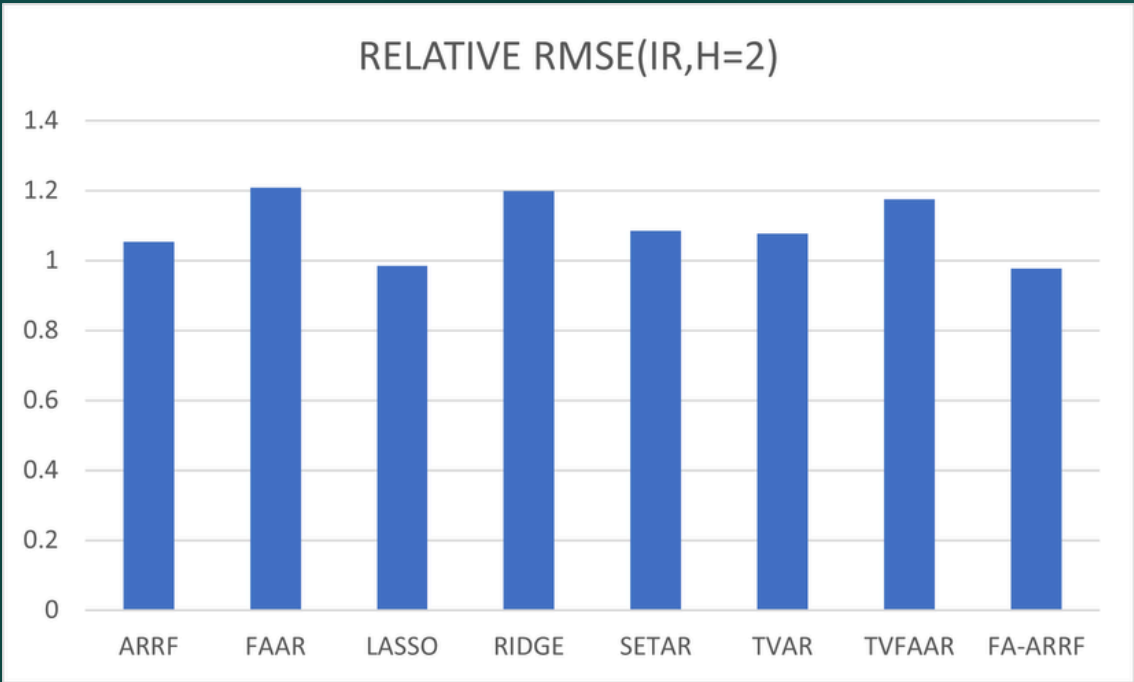
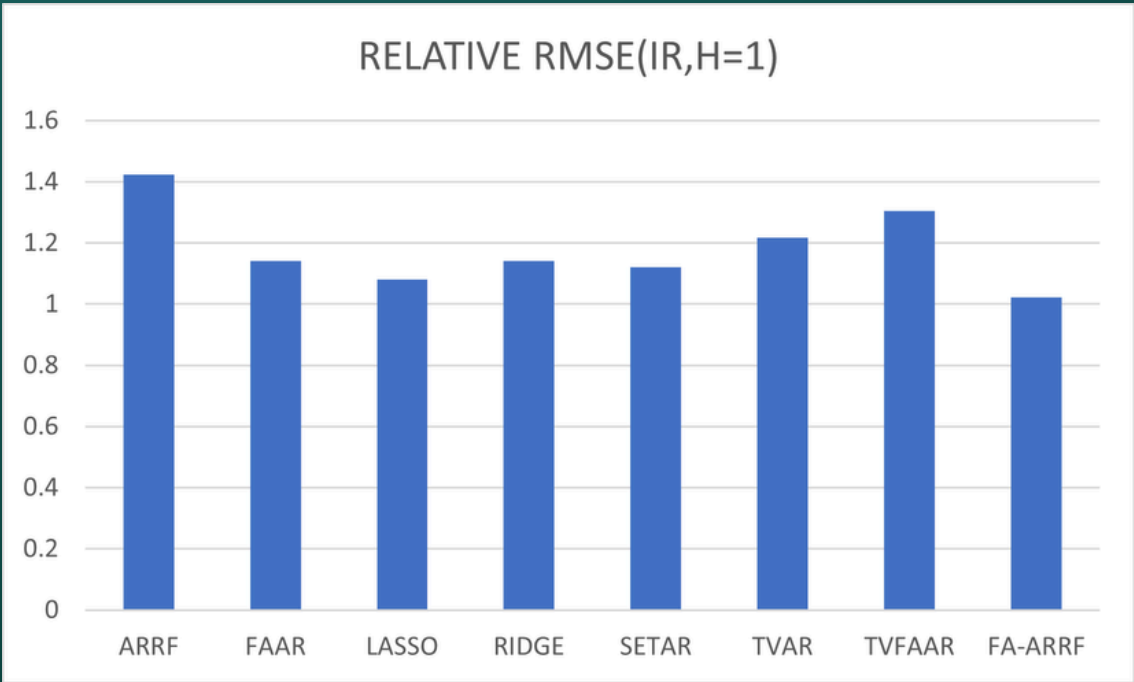
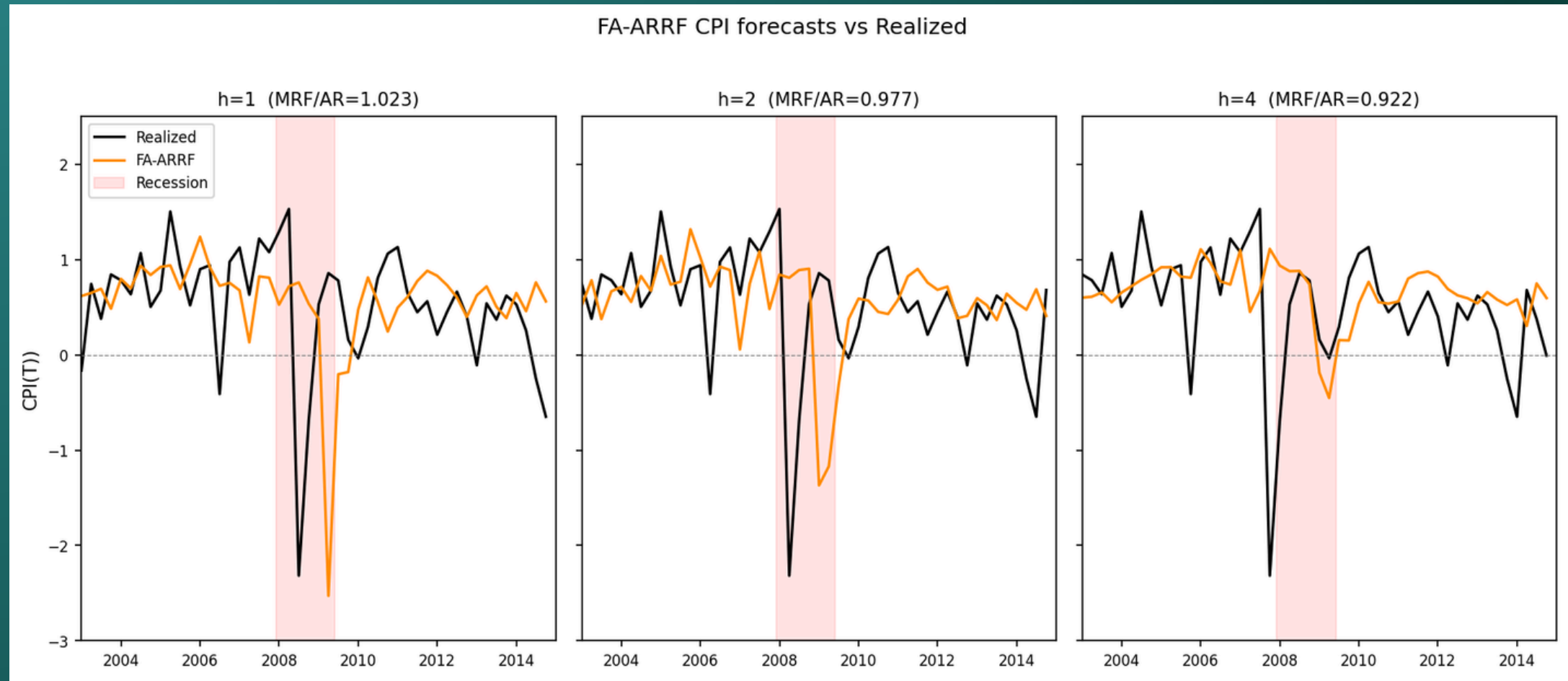
RELATIVE RMSE(GDP,H=1)



RELATIVE RMSE(GDP, H=2)



INFLATION RATE FORECASTING RESULTS



CONCLUSION

- Forecasting performance improves with nonlinear and data-rich models across GDP, unemployment, and inflation, machine learning-based models—especially FA-ARRF—consistently perform as well as or better than traditional benchmarks in most horizons.
- Linear models are insufficient in capturing macroeconomic dynamics Models like AR and FA-AR often underperform because they assume fixed relationships, which fail during periods of structural change such as the 2008 crisis.
- Nonlinear models capture regime shifts and turning points effectively FA-ARRF and related models better track sharp movements (e.g., unemployment spike, GDP collapse), indicating their ability to adapt to changing economic conditions.